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(54) **MODEL PARAMETER REDUCTIONS AND
MODEL PARAMETER SELECTION TO
OPTIMIZE EXECUTION TIME OF
RESERVOIR MANAGEMENT WORKFLOWS**

(58) **Field of Classification Search**
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(71) Applicant: **Landmark Graphics Corporation,**
Houston, TX (US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,054,752 B2 5/2006 Zabalza-Mezghani et al.
8,335,677 B2 12/2012 Yeten et al.
(Continued)

FOREIGN PATENT DOCUMENTS

JP 2004062440 2/2004
JP 2011242923 12/2011
(Continued)

OTHER PUBLICATIONS

Kassenov et al. ("Efficient Workflow for Assisted History Matching and Brownfield Design of Experiments for the Tengiz Field", 2014) (Year: 2014).*

(Continued)

Primary Examiner — Rehana Perveen

Assistant Examiner — Troy A Maust

(74) *Attorney, Agent, or Firm* — DeLizio, Peacock, Lewin & Guerra, LLP

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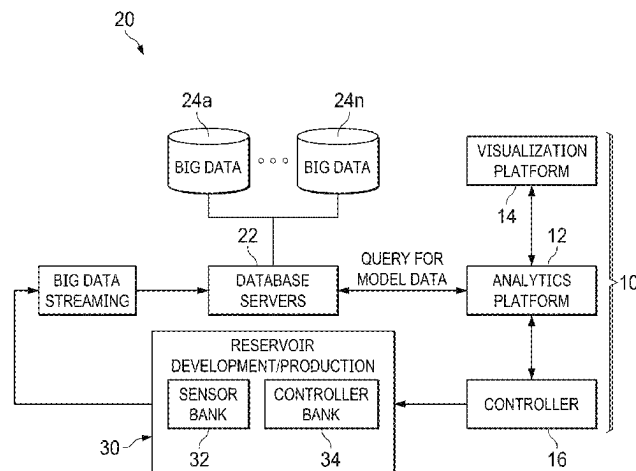
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(57) **ABSTRACT**

An apparatus for generating forecasts from a high-dimensional parameter data space comprising a reservoir model, a model order reduction module, and an assisted history matching module. The reservoir model having input variables, output variables, and an algorithmic model. The input variables, output variables, and the algorithmic model are generated by a flow simulator module and from a formation and reservoir properties database and a field production database. The model order reduction module generates a subset of the original or transformed input variables. The assisted history matching module adjust values of the output variables based on a difference between the at least one of

(Continued)



the output variables and dynamic field production data to improve model accuracy.

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(56)

References Cited

U.S. PATENT DOCUMENTS

9,151,868 B2	10/2015	Levitani
2003/0015351 A1	1/2003	Goldman et al.
2008/0133194 A1	6/2008	Klumpen et al.
2010/0155142 A1	6/2010	Thambynayagam et al.
2012/0123746 A1 *	5/2012	Postma G06F 17/13 703/2
2012/0253770 A1	10/2012	Stern et al.
2013/0246032 A1	9/2013	El-Bakry et al.

2013/0338985 A1	12/2013	Garcia et al.
2014/0039859 A1	2/2014	Marko et al.
2015/0153476 A1	6/2015	Prange et al.
2016/0055125 A1 *	2/2016	Razavi G06F 30/20 703/2
2016/0356125 A1	12/2016	Bello et al.
2018/0210977 A1	7/2018	Baddourah et al.
2022/0245300 A1 *	8/2022	Bordas G01V 99/005

FOREIGN PATENT DOCUMENTS

JP	2019179319	10/2019
KR	100566438	3/2006

OTHER PUBLICATIONS

Patel et al. ("Polynomial-Chaos-Expansion Based Integrated Dynamic Modelling Workflow for Computationally Efficient Reservoir Characterization: A Field Case Study", 2017) (Year: 2017).*

Bhark et al. ("Assisted History Matching Benchmarking: Design of Experiments-based Techniques", 2014) (Year: 2014).*

International Search Report and Written Opinion issued in related International Patent Application No. PCT/US2020/050477; mailed on Dec. 23, 2020.

International Search Report and Written Opinion issued in corresponding International Patent Application No. PCT/US2020/050478; mailed on Jan. 6, 2021.

Boxiao Li, et al.; "Best Practices of Assisted History Matching Using Design of Experiments"; SPE Journal; Aug. 2019; pp. 1435-1451.

T.E. Esmail, et al.; "Reservoir Screening and Sensitivity Analysis of Waterflooding with Smart Wells through the Application of Experimental Design"; SPE International—SPE 93568; 2005; p. 1-8.

Office Action and Search Report mailed Oct. 13, 2023 for NO Application No. 20220282.

Office Action mailed Apr. 17, 2024 for Norwegian Patent Application No. 20220282.

* cited by examiner

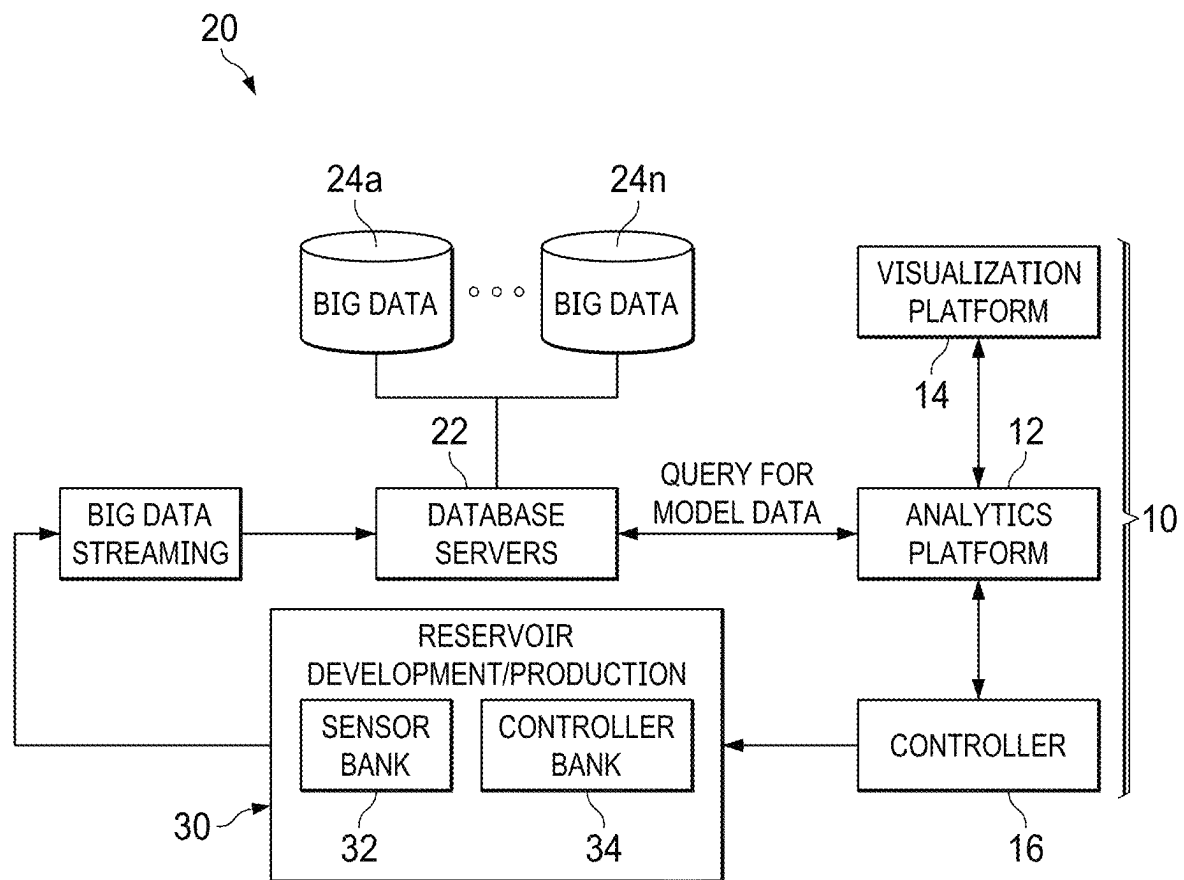
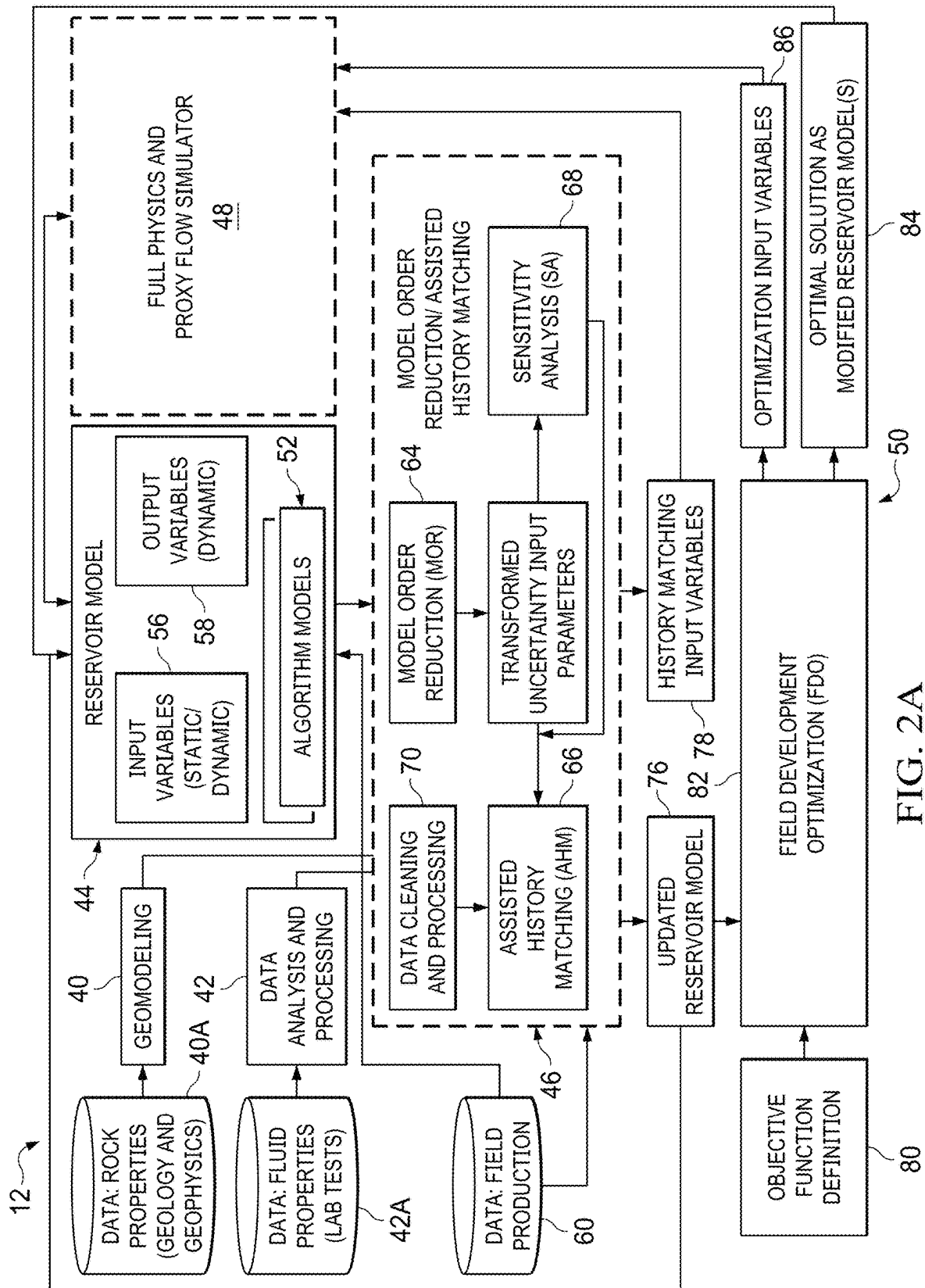


FIG. 1



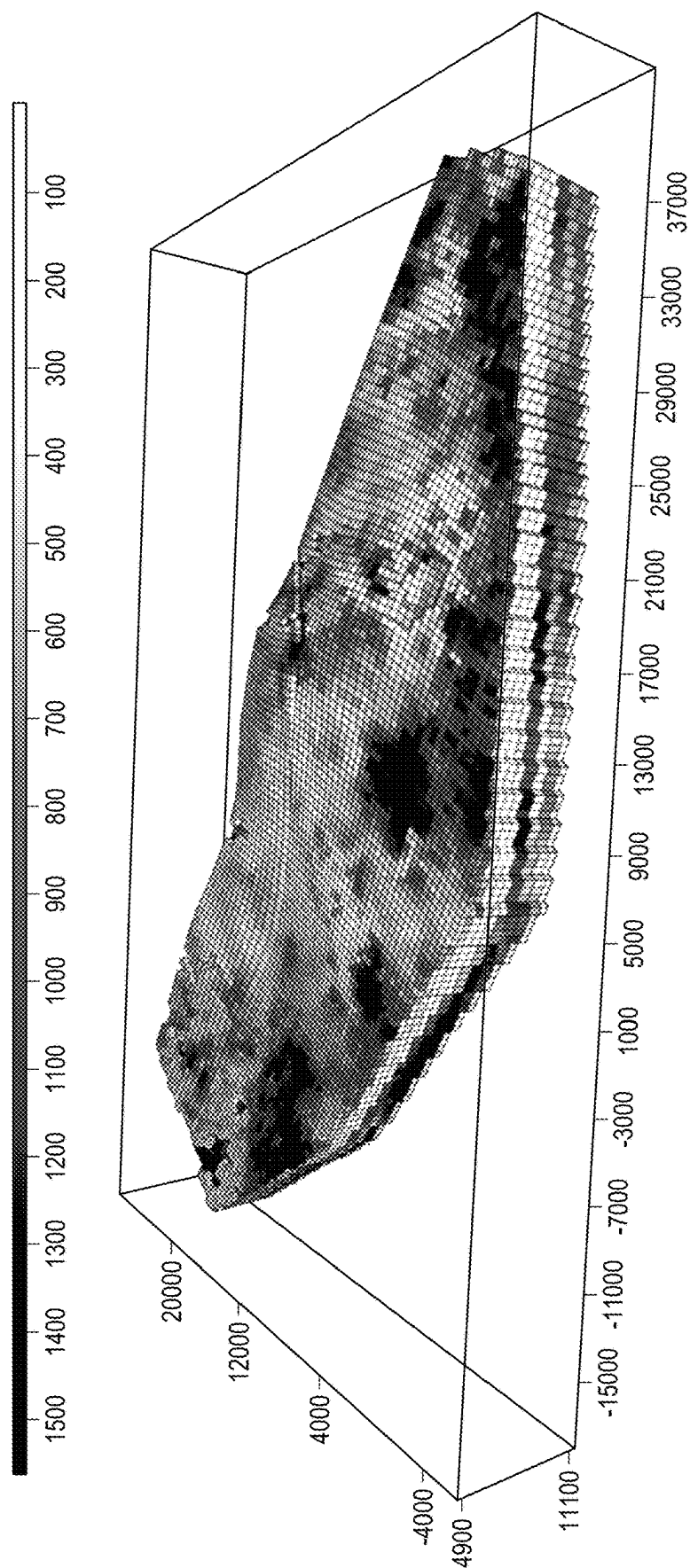


FIG. 2B

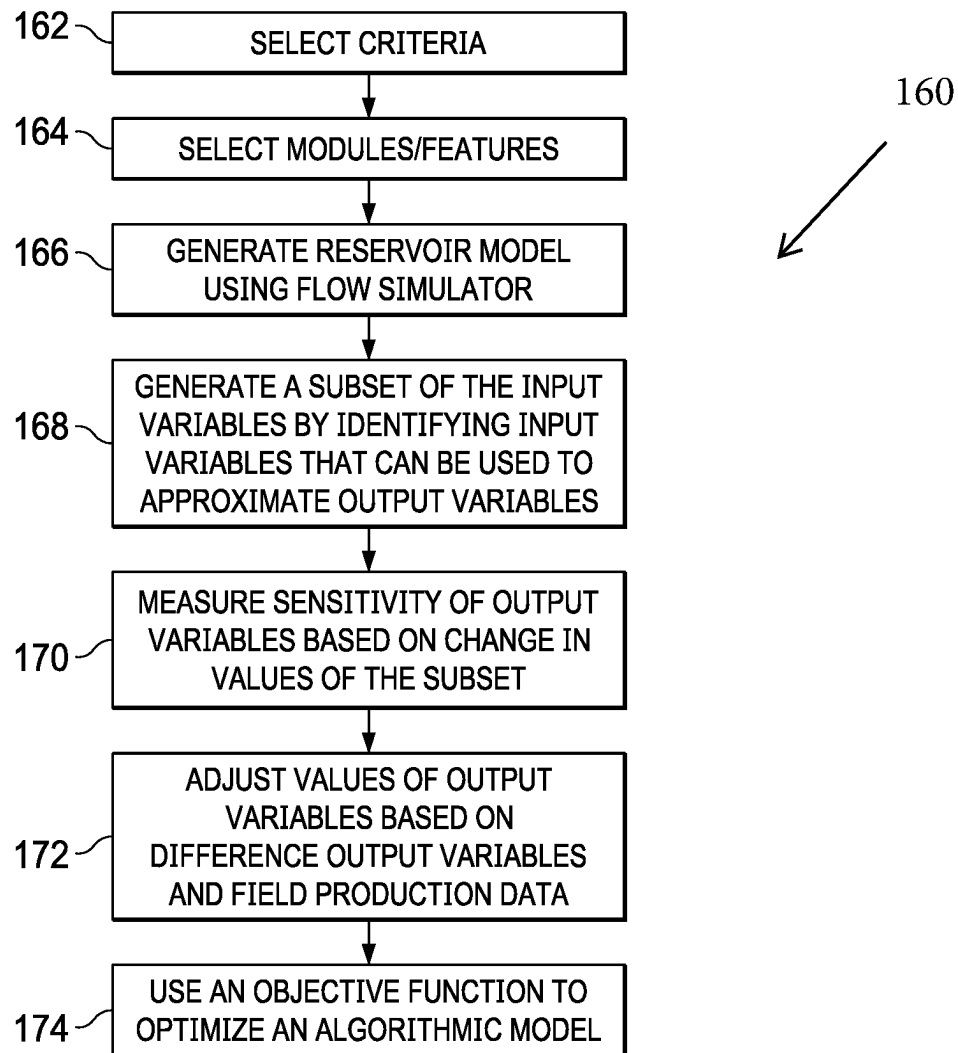


FIG. 3

FIG. 4

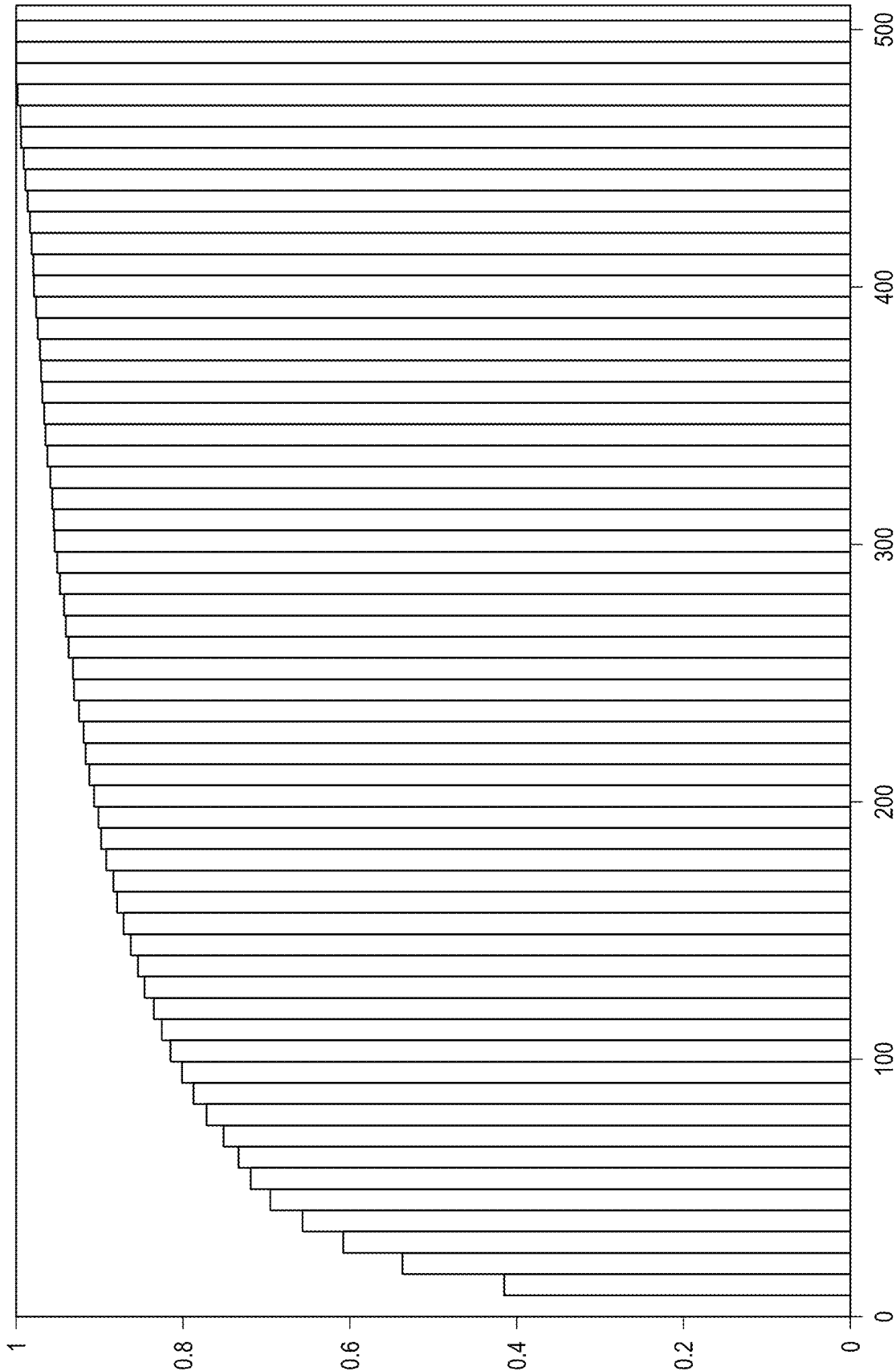
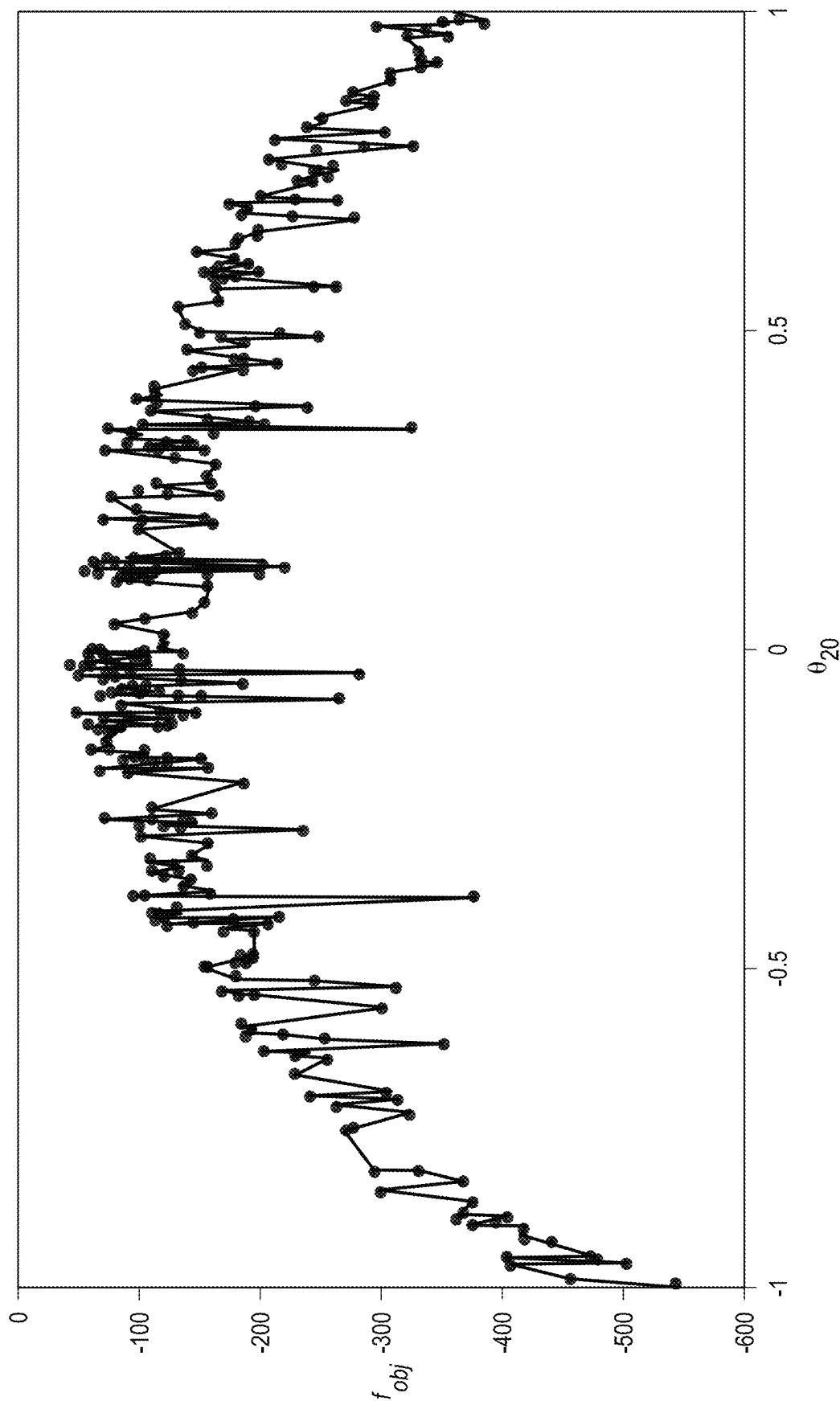


FIG. 5



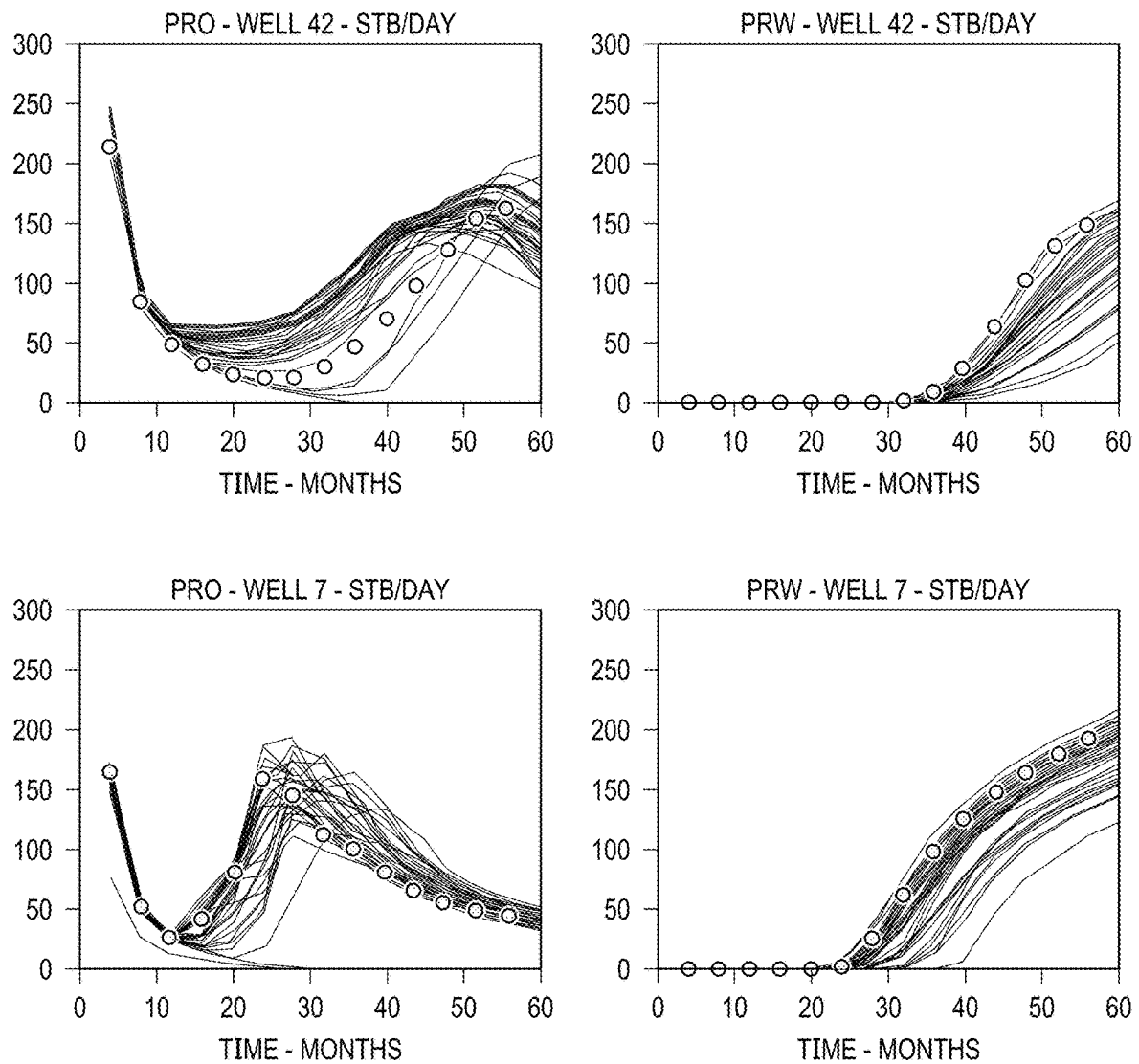


FIG. 6A

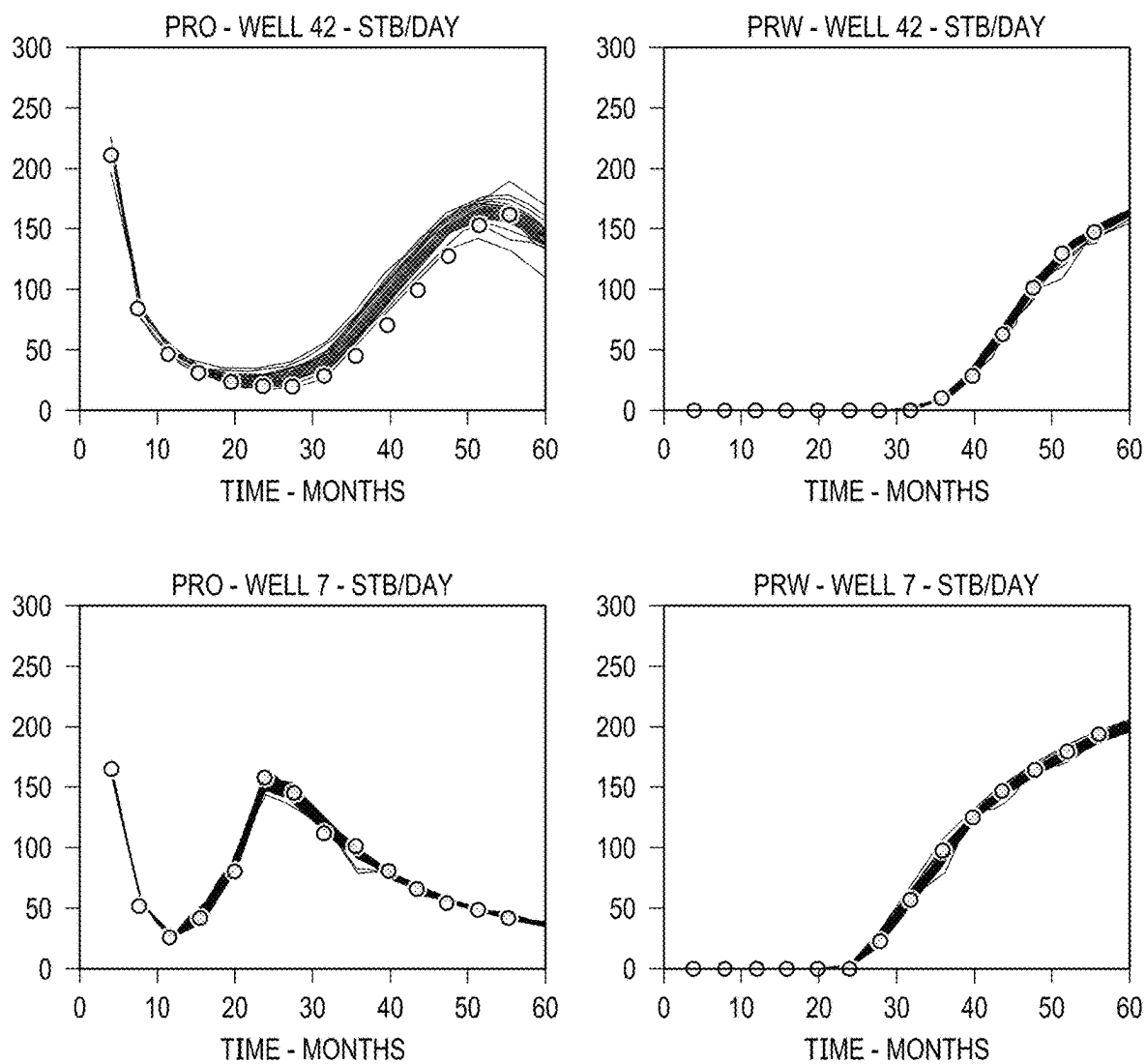
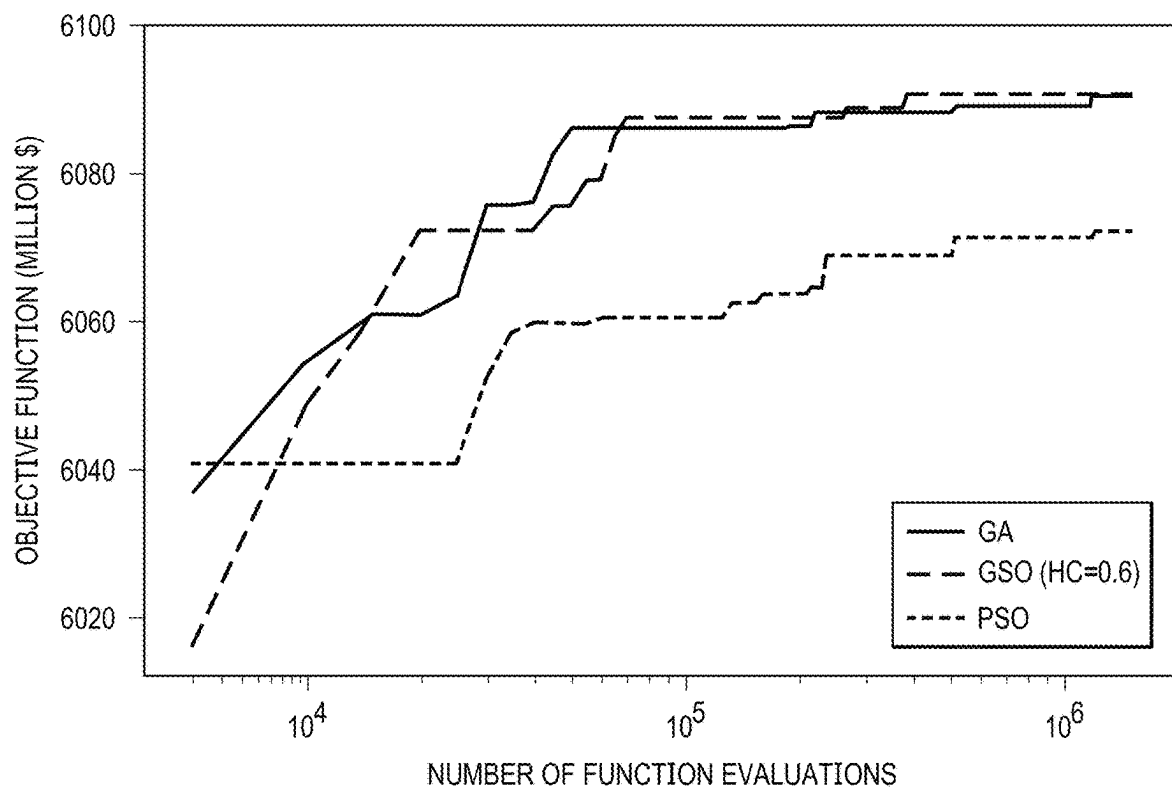


FIG. 6B

FIG. 7



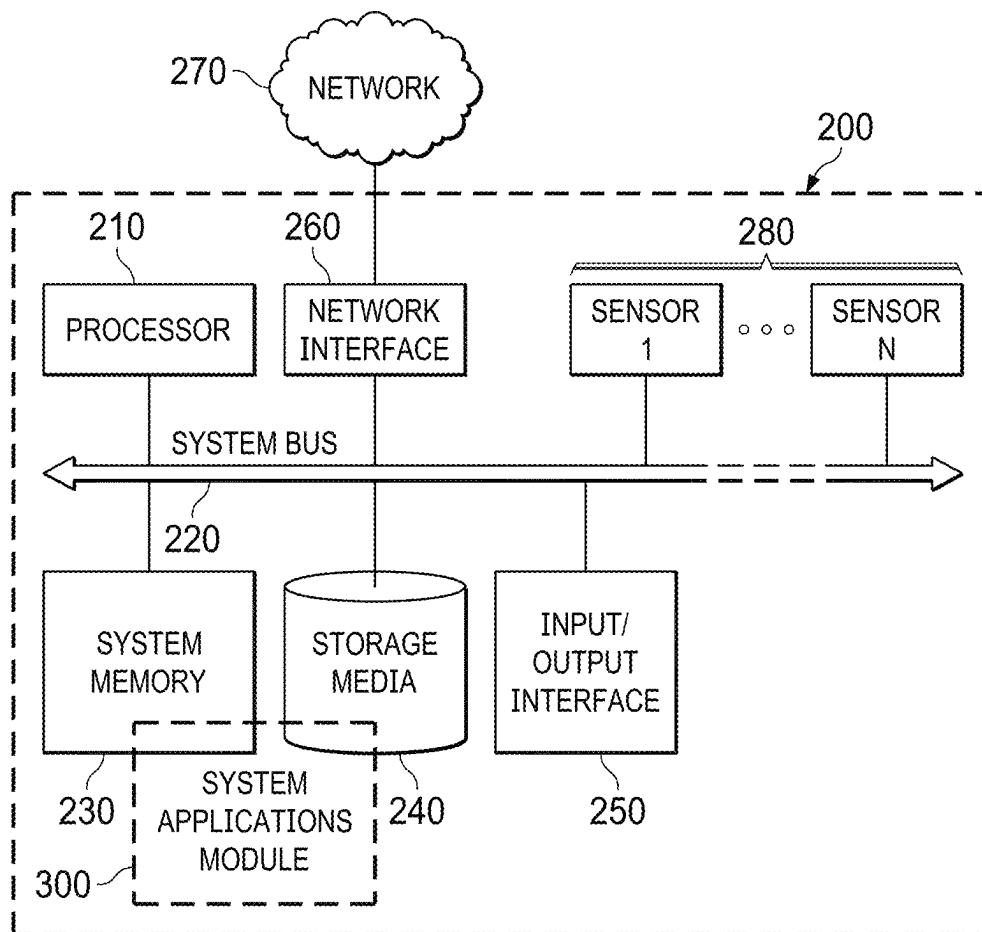


FIG. 8

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MODEL PARAMETER REDUCTIONS AND MODEL PARAMETER SELECTION TO OPTIMIZE EXECUTION TIME OF RESERVOIR MANAGEMENT WORKFLOWS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 62/929,027 filed on Oct. 31, 2019, the entire disclosure of which is expressly incorporated herein by reference.

BACKGROUND

Predictive analysis of big data based on complex workflows, i.e. algorithmic models, are computationally expensive and time consuming. However, although less computationally expensive and time consuming, less complex workflows produce less accurate forecasts, i.e. predictions. Therefore, in order to allow reservoir management teams to quickly make informed and effective decisions, there is a need for systems that can generate less computationally complex workflows yet still maintain an acceptable degree of accuracy.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the features and advantages of the present disclosure, reference is now made to the detailed description along with the accompanying figures in which corresponding numerals in the different figures refer to corresponding parts and in which:

FIG. 1 is an illustration of a diagram of a system comprising a predictive analytics tool, big data repository, and a reservoir development and/or production facility for managing reservoir development and/or production using workflows and forecasts, in accordance with certain example embodiments;

FIG. 2A is an illustration of a diagram of the integrated closed-loop analytics platform 12 that defines and generates algorithmic models used to create reservoir data models and forecasts from a high-dimensional parameter data space, in accordance with certain example embodiments;

FIG. 2B, illustrated is an example of a 3D representation of a geological structure and petrophysical properties;

FIG. 3 is an illustration of a diagram of an algorithm for a model order reduction/assisted history matching module and the flow simulator module of the integrated closed-loop analytics platform for generating reservoir management workflows and forecasts from a high-dimensional parameter data space, in accordance with certain example embodiments;

FIG. 4 is an illustration of a plot diagram for cumulative variance versus eigenvectors of two dimensional permeability field, in accordance with certain example embodiments;

FIG. 5 is an illustration of chart that plots change of an objective function used with an assisted history module with different values of a coefficient for one of the objective function's bases while other coefficients are all kept constant, in accordance with certain example embodiments;

FIGS. 6A-6B are illustrations of a graph plotting results of assisted history matching and Bayesian optimization, in accordance with certain example embodiments;

FIG. 7 is an illustrations of a graph showing optimization results for an objective function, such as particle swarm optimization (PSO), genetic algorithm (GA), or a genetical

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swarm optimization (GSO) schemes, with an optimization module, in accordance with certain example embodiments; and

FIG. 8 is an illustration of a diagram of a computing machine and a system applications module 300, in accordance with certain example embodiments.

DETAILED DESCRIPTION

While the making and using of various embodiments of the present disclosure are discussed in detail below, it should be appreciated that the present disclosure provides many applicable inventive concepts, which can be embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative and do not delimit the scope of the present disclosure. In the interest of clarity, not all features of an actual implementation may be described in the present disclosure. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming but would be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.

Traditional hydrocarbon reservoir workflows used to generate reservoir characterization data models are usually generated using high dimensional parameter space data models and full physics algorithmic model or models. This method of generating data models requires a significant amount of computational resources and computational time during reservoir simulation. As such, decision making and planning in the reservoir management process can be delayed. This delay can introduce unwanted expenses and costs for well site developers and operators.

Presented herein is a system, method, and apparatus that uses a flow simulator and at least one selected from a group comprising model-order reduction techniques, history matching techniques, and optimization techniques to generate workflows with each subsequently generated workflow having a reduced parameter space than a previously generated workflow. As such, select workflows are less computationally expensive and computationally time consuming yet produce accurate, reliable forecasts. The forecasts can be used to aid in the decision making process in the management of hydrocarbon reservoir development and production.

Algorithmic models are generated using a flow simulator. The flow simulator generates output variables using input variables from a formation and reservoir properties database, a field production database, and input variables from at least one selected from a group comprising model order reduction techniques, history matching techniques, and optimization techniques. Model order reduction techniques are used to generate a subset of the input variables, the subset having a reduced parameter space, by identifying input variables that can be used to approximate the output variables. Assisted history matching techniques are used to adjust values of the output variables based on a difference between the at least one of the output variables and dynamic field production data. In some embodiments, a function decomposition algorithm or a design of experiments algorithm can be used to identify input variables that can be used to approximate the output variables. The input variables include reservoir parameters represented by a coefficient and the subset is a subset of the reservoir parameters represented

by another coefficient that has a value less than that of the coefficient. In some embodiments, the other coefficient is derived by applying a series expansion algorithm to the input variables and the output variables. In some embodiments, sensitivity analysis techniques are used to measure sensitivity of output variables based on change in values of the subset of the reservoir parameters. In some embodiments, Bayesian optimization can be used to adjust values of the output variables based on a difference between the output variables and dynamic field production data.

In this specification, workflow means an algorithmic model or models and an associated data model or models. Geological data model means a high-dimensional parameter data space of reservoir properties, i.e. permeability and porosity. Big data means a high-dimensional parameter data space, e.g. millions of rows and/or columns of variables. Machine learning algorithms can mean, as an example, simple multivariate regression model (MVR), artificial neural network (ANN), and a decision tree-based algorithm named extreme gradient boosting (XGB). These algorithms are compared for construction of a proxy model. Optimization algorithm can mean, as an example, particle swarm optimization (PSO), genetic algorithm (GA), and a hybrid approach of PSO and GA, called genetical swarm optimization (GSO). Uncertainty variables are related to the uncertainty quantification and the science of quantitative characterization. Output variables, predicted variable, forecast variables are used interchangeable within this specification. Reduced parameter space means the number of parameters associated with an algorithmic model and input variables have a reduced number of parameters compared with an original algorithm model, or a previously generated algorithmic model, and input variables associated with the algorithmic model, or the previously generated algorithm model.

Referring now to FIG. 1, illustrated is a diagram of a system comprising a predictive analytics tool 10, big data repository 20, and a reservoir development and/or production facility 30 for managing reservoir development and/or production using workflows and forecasts, according to certain example embodiments. The analytics tool 10 comprises an integrated closed-loop analytics platform 12, a visualization platform 14, and a controller 16. The integrated closed-loop analytics platform 12 is communicable coupled with the big data repository 20, the visualization platform 14, and the reservoir production and/or development facility 30 through the controller 16. The big data repository 20 comprises database servers 22 and storage banks 24a-24n for receiving high velocity, high volume streaming data or data otherwise communicated periodically from the reservoir development and/or production facility 30. The reservoir development and/or production facility 30 comprises a sensor bank 32 and a controller bank 34. The sensor bank 32 is communicable coupled to the database base servers 22 and can stream data from the facility 30 or load the data periodically. The controller bank 34 is communicable coupled with the controller 16. Based on data models queried from the big data repository 20, generated algorithmic models, and other relevant information, forecasts, i.e. predictions, the integrated closed-loop analytics platform 12 can generate visualizations and/or generate control signals to control the controller bank 34 through the controller 16.

Referring now to FIG. 2A, illustrated is a diagram of the integrated closed-loop analytics platform 12 that defines and generates algorithmic models used to create reservoir data models and forecasts from a high-dimensional parameter data space, according to certain example embodiments. The

analytics platform 12 comprises a geomodeling module 40, a data analysis module 42, a reservoir model 44, a model order reduction/assisted history matching module 46, a flow simulator module 48, and an optimization module 50. The analytics platform 12 can use a proxy flow simulator, trained algorithmic models, model order reduction techniques, assisted history matching techniques, and an optimization process to supplement a full-physics flow simulator to generate algorithmic models that are less computationally complex and, therefore, more efficient. These algorithmic models can be used against a high-dimensional parameter data space, commonly referred to as Big Data, to generate reservoir data models with predictive results. It should be understood by those of skill in the art that the analytics platform 12 is a looped computational system that continuously process the reservoir model 44 until a predefined criteria is satisfied. Furthermore, it should also be understood a user may elect not to include all features or sub-features of the analytics platform 12. As an example, the user may choose to use select features from the flow simulator 48, e.g. a proxy flow simulator, choose not to use the optimization module 50, and only certain features of the model order reduction/assisted history matching module 46.

The geomodeling module 40 generates static input variables from geology and geophysics database 40A. The static input variables can include geological data, such as formation structure derived from geophysical interpretations, and petrophysical rock properties, such as lithological facies, porosity, and absolute permeability attributes for a particular earth formation and hydrocarbon reservoir. The data analysis module 42 also generates static input variables from lab tests database 42A. The static input variables can include fluid and rock property variables from lab measurements based on a fluid and rock type or types. Stated differently, the static input variables are geological data and petrophysical data produced on different types of rocks and fluids. The variables describe physical properties of the fluid and rock, such as compressibility, relative permeability, thermodynamic interaction, and more.

The reservoir model 44 can comprise algorithmic models 52, a data pool of variables, and, optionally, a 3-Dimensional (3D) representation of the geological structure and petrophysical properties of a corresponding earth formation and hydrocarbon reservoir using the data pool of variables. The algorithmic models 52 and data pool of variables evolve over an iterative computation process based on predefined criteria. The algorithmic models 52 comprise different models having different degrees of complexity. The algorithmic models 52 are produced over the processing cycles of the analytics platform 12 and include an algorithmic model that satisfies the predefined criteria. The data pool of variables comprises input variables 56 from the geomodeling module 40, the data analysis module 42, and a field production database 60. The input variables 56 can be static and/or dynamic. The geological properties are assumed static whereas the field production and geophysical data from the field production database 60 and geophysics database 40A may be dynamic and/or static. The data pool of variables also comprises dynamic output variables 58 generated by the flow simulator module 48, the model order reduction/assisted history matching module 46, and the optimization module 50 based on the input variables 56 and previously processed output variables 58. The dynamic output variables 58 are predicted, i.e. forecasted, variable values, such as pressure and fluid (oil, water, gas) saturation on a grid or fluid production rates at the well locations. Initially, the input variables 56 can be considered of a large data, wherein

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the input variables 56 can comprise millions of rows and/or columns of data, and the algorithmic model has a high order of parameters. However, the number of input variables 56 and number of algorithmic parameters are reduced through iterative processing using the model order reduction/assisted history matching module 46. The predefined criteria can be based on an optimal solution, an amount of processing time, a final algorithmic model having a given parameter size, or an acceptable algorithmic model accuracy.

It should be understood that the term static and dynamic used to describe input variables 56 and output variables 58 are in relation to time as it relates to Earth formation and reservoir properties. However, as just stated, the number of variables can change and in some cases the values of some variables may change due to approximation of calculations and other functions performed by some features of the analytics platform 12. It should also be understood that the reservoir model 44 can be generated and optimized efficiently and effectively using the flow simulator module 48, the model order reduction/assisted history matching module 46, and the optimization module 50. However, a user can select which features or sub features of the flow simulator 48, the model order reduction/assisted history matching module 46, and the optimization module 50 are used to generate the reservoir model 44.

The 3D representation can be generated to include a grid. Geostatistical techniques can be used to populate the grid using the variables describing the geological and petrophysical rock properties with correct spatial continuity. Grid block or blocks can be populated using the variables describing the fluid and rock properties generated from lab tests. The grid can also be populated with data from the field production database 60, such as operating bottomhole pressure or injection water rate. This data is also considered as input variables to the flow simulator module 48. FIG. 2B is an example of a 3D representation of a geological structure and petrophysical properties. In addition, the user can interact with the 3D representation to change certain variables, e.g., change the size of the modeled Earth formation, the field production variables, and/or the predicted variables. Using the flow simulator module 48, the model order reduction/assisted history matching module 46, and the optimization module 50 regeneration of the 3D representation is also efficient and effective.

In an embodiment, the flow simulator module 48 can generate an initial reservoir model using a full-physics flow simulator, e.g. such as the simulator that can be found in Nexus® software from Halliburton Landmark, and iteratively reduce the parameter space of the initial reservoir model based on the input variables 56, output variables 58, and output from at least one selected from a group comprising the model order reduction/assisted history matching module 46 and the field development optimization module 50. In another embodiment, the flow simulator module 48 can generate the initial reservoir model using the full-physics flow simulator and iteratively reduce the parameter space of the initial reservoir model based on the input variables 56, output variables 58, and output from at least one selected from a group comprising the model order reduction/assisted history matching module 46, the field development optimization module 50, a proxy flow simulator for assisted history matching module, a proxy flow simulator for field development optimization module, a design of experiments module, and a machine learning and validation module.

The model order reduction/assisted history matching module 46 comprises at least one selected from a group

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comprising a model order reduction module 64, an assisted history matching module 66, a sensitivity analysis module 68, and a data cleaning and processing module 70. The model order reduction/assisted history matching module 46 can generate a reduced list of history matching input variables 78 of the reservoir model 44 and an updated reservoir model 76 having a reduced parameter space. The reduced list of history matching input variables 78 is a list of input variables 56 wherein that list of variables has a reduced parameter space than that of input variables 56 initially processed by the module 46. Using one or both of the history matching input variables 78 and the updated reservoir model 76, the flow simulator module 48 can generate predictions more efficiently. Depending on the features used, the flow simulator module 48 can generate the predictions more efficiently and further reduce the parameter space of the reservoir model 44.

The model order reduction module 64 uses a function decomposition or design of experiments algorithm to reduce the parameter space of the reservoir model 44 by identifying input variables 56 that can be used to best approximate the output variables 58. In other words, the number of input variables 56, as determined by the flow simulator module 48, used by the algorithmic model to determine output variables 58 is reduced using function decomposition or a design of experiments algorithm to identify a subset of input variables 58 that can be used to simulate a similar predictive result or results. The identified variables are transformed uncertainty input variables. The transformed uncertainty input variables are a reduced parameter space version of the input variables 56. The input variables 56 can be a large number of gridded reservoir parameters, such as porosity or permeability distributed on a grid, represented by a large coefficient. This large number of gridded reservoir parameters can be represented by a smaller number of coefficients. These new coefficients define the reduced parameter space version of the uncertainty input variables. The new coefficients can be derived by applying a series expansion algorithm, such as Karhunen-Loeve expansion, to the input variables 56 and output variables 58. The sensitivity analysis module 68 screens the input variables 56 and the transformed uncertainty input variables. The sensitivity analysis module 68 examines how output variables 58, identified for history matching, are sensitive to the change in values of the transformed uncertainty input variables. The most influential uncertainty input parameters are selected and kept in the history matching input variables 78.

The history matching module 66 uses optimization techniques to update reservoir model 44 using dynamic field production data 60, such as oil, water, and gas production rates. Using an optimization technique, such as Bayesian optimization, the history matching module can reduce the mismatch between reservoir model predictions and observed production data 60. In essence, the history matching module 46 corrects the output variables 58 based on the difference between the field production data 60 and the output variables 58. In particular, values of the selected uncertainty input parameters are adjusted to account for actual values of production data 60. The updated reservoir model 76 can be used for future predictions by running either full-physics flow simulations for future time steps or by using proxy flow model to get forecasted values of output variables 58. The optimization module 50 can use the updated reservoir model 76 to define a best case scenario or scenarios to manage a reservoir by, e.g., finding optimal operation settings or by defining optimal locations for new well placement.

The optimization module **50** comprises an objective function definition module **80** and a field development optimization module **82**. The objective function definition module **80** and the field development optimization module **82** generates an optimized reservoir model **84** to identify best field development scenario or scenarios and optimization input variables **86** to construct a proxy flow model for the optimization module **50**. The field development optimization module **82** is used to define a best case scenario or scenarios to operate a well development and production site by finding optimal operation settings and/or by defining optimal locations for new well placement. This procedure for finding optimal reservoir management scenarios is solved with an optimization technique, such as particle swarm optimization or genetic algorithm. The objective function definition module **80** defines a particular objection function or functions. The function or functions provide an analytical form for optimality of the optimization problem, such as net present value (NPV) of a hydrocarbon reservoir project. Different scenarios can be generated by varying values of the optimization input parameters that are different from the uncertainty input parameters. An example of the optimization input parameters is oil and gas production rate at the production well, water injection rate at the injection well, locations of the wells, locations of the valves, and so on; in other words, the parameters of the reservoir equipment settings, values of which can be changed by a human operator manually or automatically. The optimal solution is represented by one or several modified reservoir models **84**.

Society of Petroleum Engineers (SPE) 93568, "Reservoir Screening and Sensitivity Analysis of Waterflooding With Smart Wells Through the Application of Experimental Design", T. E. H. Esmail, Delft U. of Technology, Kuwait Inst. For Scientific Research; S. Fallah, Delft U. of Technology, Natl. Iranian Oil Co.; and C. P. J. W. van Kruijsdijk, Delft U. of Technology, Copyright © 2005, Society of Petroleum Engineers Inc., is an example of white paper that discusses the use of experimental design methods to find an optimal proxy model in a more computationally efficient manner. "Best Practices of Assisted History Matching Using Design of Experiments," Boxiao Li, Chevron Energy Technology Company; Eric W. Bhark, Chevron Asia Pacific E&P Company; and Stephen J. Gross (ret.), Travis C. Billiter, and Kaveh Dehghani, Chevron Energy Technology Company, August 2019 SPE Journal, Copyright © 2019 Society of Petroleum Engineers, is another white paper that discusses the use of experimental design methods and assisted history matching to find an optimal proxy model in a more computationally efficient manner.

Referring now to FIG. 3, illustrated is a diagram of an algorithm **160** for the model order reduction/assisted history matching module **46** and the flow simulator module **48** of the analytics platform **12** for generating reservoir management workflows and forecasts from a high-dimensional parameter data space, according to certain example embodiments. The algorithm **160** generates output variables **58** using algorithmic models **52** generated using selected modules and features of the flow simulator module **48**, model order reduction/assisted history matching module **46**, and field development optimization module **50**. The algorithmic models **52** are generated until the predefined criteria is satisfied. The last model generated is capable of generating a data model with predictive variables (the output variables **58**) using the input variables **56** or a subset of the input variables **56** in a computationally efficient manner. As previously stated, the predefined criteria can be based on an optimal

solution, an amount of processing time, a final algorithmic model having a given parameter size, or an acceptable algorithmic model accuracy.

The algorithm **160** begins at block **162** where a criteria (objective/goal) for generating an algorithmic model **162** is selected. For example, an optimal field development scenario or scenarios can be identified for a petroleum reservoir under study. At block **164**, the algorithm **160** selects the modules **46**, **48**, and **50** used in generating the algorithmic models **52** and the features, e.g. full-physics flow simulator, a proxy flow simulator, assisted history matching module **66**, model order reduction module **64**, etc. At block **166**, the algorithm **160** generates the reservoir model **44**. Initially, the reservoir model **44** only includes the input variables **56**. The input variables **56** are a product of the geomodeling module **40**, the data analysis module **42**, and field production database **60**. A subset of the input variables **56**, such as petrophysical properties of a petroleum reservoir, can be generated by applying geostatistical techniques, such as kriging interpolation technique or Sequential Gaussian Simulation (SGS), to well data. Other algorithms are used jointly with structural data, and seismic data, or a combination of those to generate other static properties of the reservoir, such as reservoir model grid definition. Stated differently, the input variables **56** can be conditioned to sub-surface logging tool input data and/or geological survey data, such as well data, structural data, and seismic data, or a combination of those. The values of input variables **56** are generated to represent the uncertainty and variability in the model parameters. The input variables **56** can include porosity variables and permeability variables, as an example. After the input variables **56** are generated, an initial algorithmic model and output variables **58** are generated using a full-physics flow simulator module. Subsequent algorithmic models can be generated continuously using the initial algorithmic model and the features selected until the criteria is satisfied. As an example, the parameter space of the initial algorithmic model can be reduced in subsequent processing by using a proxy flow simulator and trained algorithmic models with input provided from the model order reduction/assisted history matching module **46** and optimization module **50**.

As an example, the algorithmic models can be trained using the initial algorithmic model, a plurality of training and validation datasets, and the reservoir model **44** and any updates thereto that are realized during the processing cycle. As another example, design of experiments can be used to reduce the amount of time required to generate the trained algorithmic models. In yet another example, the algorithmic models can be trained using the initial algorithmic model, a plurality of training and validation datasets, the reservoir model **44** and any updates thereto that are realized during the processing cycle, and history matching input variables **78**. In still yet another example, the algorithmic models can be trained using the initial algorithmic model, a plurality of training and validation datasets, the reservoir model **44** and any updates thereto that are realized during the processing cycle, and optimal input variables **86**. In yet another example, the algorithmic models can be trained using the initial algorithmic model, a plurality of training and validation datasets, the reservoir model **44** and any updates thereto that are realized during the processing cycle, history matching input variables **78**, and the optimization input variables **86**. It should be understood that the trained algorithmic models are the same as algorithmic models **52**. Machine learning algorithms, such as multivariate regression model (MVR), artificial neural network (ANN), or a decision tree-based algorithm, or any combination thereof, can be

used to generate the training and validation datasets and the trained algorithmic models. The decision tree-based algorithm can be an extreme gradient boosting (XGB) algorithm.

At block 168, the algorithm 160 processes the reservoir model 44 by generating a subset of the input variables 56, the subset having a reduced parameter space, by identifying input variables that can be used to approximate at least one output variable 58. The algorithm 160 can use a function decomposition algorithm or a design of experiments algorithm to identify input variables 56 that can be used to approximate output variables 58. The input variables 56 can include reservoir parameters represented by a coefficient and the subset is a subset of the reservoir parameters represented by another coefficient having a value less than that of the coefficient. The algorithm 160 derives the other coefficient by applying a series expansion algorithm to the input variables 56 and the output variables 58.

An example of a function decomposition algorithm is Karhunen-Loeve expansion. A Karhunen-Loeve expansion equation can be used to identify at least one heterogeneous group of parameters. Specifically, the number of variable parameters can be reduced by identifying a heterogeneous group of variable parameters using:

$$Y(x) = \Phi \sqrt{\lambda} \theta + \bar{Y}(x); \text{ where}$$

- Y: petrophysical property, e.g. porosity, permeability;
- x: the spatial coordinate;
- Φ : truncated eigenvector matrix;
- λ : truncated eigenvalue matrix;
- θ : vector of random variables, $\theta \sim N(0, 1)$; and
- $\bar{Y}$: average petrophysical property of the ensemble.

The reduction of parameters can be realized by truncating the eigenvector matrix based on a cumulative variability preserved, as illustrated in FIG. 4, which shows the result for a 2D test case, which is a layer of a test reservoir, i.e. test geological data model. The x-axis is the index of eigenvectors, and the y-axis is the cumulative variability, i.e. the cumulative sum of eigenvalues divided by the sum of all eigenvalues. For instance, if only the first 25 eigenvectors are kept, the preserved variability is 55% for the 2D case. Therefore, for the 3D case, more eigenvectors need to be preserved to maintain the same accuracy as the 2D case. The corresponding coefficients θ are history matching parameters. The eigenvector and eigenvalue matrix of permeability field are calculated by performing single value decomposition to the covariance matrix estimated from the permeability model ensemble. The eigenvectors, a.k.a. bases, can be visualized by reshaping them to the reservoir dimension. All history matching and optimization operations related to properties such as permeability is done not on permeability values directly but on the transformed Karhunen-Loeve expansion parameters represented by random variables θ . In such way, instead of operating on 8,000 permeability values, for example, for a layer of a geological data model, 50 random variables can be used after expansion and truncation. The reservoir model 44 is 160 times less in size than original model. FIG. 4 illustrates that in order to preserve 80% of the permeability field accuracy (spatial variability), about 100 values of the decomposed permeability field are needed to be kept, which is 25 times less than a total number of the original permeability values being at about 2,500 values. It is obvious that KLE effectively reduces the parameter space of the reservoir model 44.

At block 170, the algorithm 160 measures sensitivity of the output variable 58 based on change in values of the subset of the reservoir parameters. Sensitivity analysis can be carried out by utilizing fractional factorial design and

showing results in the form of the tornado chart. The selected input variables are varied one at a time between their distribution ends, e.g. between p10 and p90, while values of the remaining input variables are fixed at their median or mean values. The corresponding values of the selected output variable are recorded. By doing this procedure for each input variable, the output variable response is recorded and is represented by a variation range. The results are plotted as bars, which show variation range of output variable, in a decreasing order with the names of the input variables that caused such variation next to the bars. Finally, a threshold is applied to the variation range of the output variable to screen the most influential input variables. The input variables that lead to a larger variation of the output variable are considered the more influential and should be included in history matching and/or optimization blocks based on the user specified threshold.

At block 172, the algorithm 160 adjusts values of the output variables 58 based on a difference between the at least one of the output variables and dynamic field production data. As an example, Bayesian optimization can be used to adjust the values. A part of using Bayesian Optimization is to find an appropriate acquisition function suitable for a particular problem. FIG. 5 plots the change of the objective function with different values of the coefficient θ_{20} for one of the bases while the other coefficients are all kept constant. It can be noticed that despite visible trend of the objective function in the form of an arc, the local fluctuations are prominent. For this reason, the upper confidence bound (UCB) can be selected as the acquisition function to ensure that the function is more stable locally. Multithreading can also be implemented to further improve the efficiency. For example, each of the θ_i values can have a range of $(-1, 1)$. By splitting the range into multiple intervals, multiple Bayesian optimization processes can run concurrently. The acquisition function with the highest objective function can be selected from the results. The results of history matching and Bayesian Optimization are illustrated in FIGS. 6A-6B. The results illustrate the production curves (PRO—oil production rate, and PRW—water production rate) before, see FIG. 6A, and after, see FIG. 6B, history matching and Bayesian Optimization for two randomly chosen wells. A better match is obtained through the history matching with the Bayesian optimization.

At block 174, the algorithm 160 identifies and selects certain output variables 58 for optimization. The output variables 58 can be identified and selected based on a defined reservoir process, such as water treatment and water injection costs. The output variables can be optimized by comparing output variables generated using an objective function and generated using an algorithmic model. The algorithmic proxy model can be optimized using:

$$f_{obj} = p_o c_o + p_g c_g - p_w c_w - p_{iw} c_{iw};$$

- where, p_o and p_g are oil and gas prices;
- p_w and p_{iw} are the costs of water treatment and water injection, respectively;
- c_o , c_g , and c_w are cumulative oil, gas and water produced;
- c_{iw} is the cumulative water injected.

The objective function, f_{obj} , can be a particle swarm optimization (PSO), genetic algorithm (GA), or a genetical swarm optimization (GSO). As an example, the predictive outcome variables can be field water, oil, and gas production rates. The proxy flow model is used to make water production rate, oil production rate, gas production rate, and water injection rate predictions. FIG. 7 illustrates the optimization results for the GA, GSO, and PSO schemes. For this study,

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GA and GSO outperform PSO in terms of the optimum value reached and the rate at which better solutions are found. Note that the objective of this study is not to generalize the relative performances of the three optimization methods, but rather to show their applicability for the intended purpose.

Referring now to FIG. 8, illustrated is a computing machine 200 and a system applications module 300, in accordance with example embodiments. The computing machine 300 can correspond to any of the various computers, mobile devices, laptop computers, servers, embedded systems, or computing systems presented herein. The module 300 can comprise one or more hardware or software elements designed to facilitate the computing machine 200 in performing the various methods and processing functions presented herein. The computing machine 200 can include various internal or attached components such as a processor 210, system bus 220, system memory 230, storage media 240, input/output interface 250, a network interface 260 for communicating with a network 270, e.g. a loopback, local network, wide-area network, cellular/GPS, Bluetooth, WIFI, and WIMAX, and sensors 280.

The computing machine 200 can be implemented as a conventional computer system, an embedded controller, a laptop, a server, a mobile device, a smartphone, a wearable computer, a customized machine, any other hardware platform, or any combination or multiplicity thereof. The computing machine 200 and associated logic and modules can be a distributed system configured to function using multiple computing machines interconnected via a data network and/or bus system.

The processor 210 can be designed to execute code instructions in order to perform the operations and functionality described herein, manage request flow and address mappings, and to perform calculations and generate commands. The processor 210 can be configured to monitor and control the operation of the components in the computing machines. The processor 210 can be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor ("DSP"), an application specific integrated circuit ("ASIC"), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. The processor 210 can be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. According to certain embodiments, the processor 210 along with other components of the computing machine 200 can be a software based or hardware based virtualized computing machine executing within one or more other computing machines.

The system memory 230 can include non-volatile memories such as read-only memory ("ROM"), programmable read-only memory ("PROM"), erasable programmable read-only memory ("EPROM"), flash memory, or any other device capable of storing program instructions or data with or without applied power. The system memory 230 can also include volatile memories such as random access memory ("RAM"), static random access memory ("SRAM"), dynamic random access memory ("DRAM"), and synchronous dynamic random access memory ("SDRAM"). Other types of RAM also can be used to implement the system memory 230. The system memory 230 can be implemented using a single memory module or multiple memory modules. While the system memory 230 is depicted as being part of the computing machine, one skilled in the art will recognize that the system memory 230 can be separate from

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the computing machine 200 without departing from the scope of the subject technology. It should also be appreciated that the system memory 230 can include, or operate in conjunction with, a non-volatile storage device such as the storage media 240.

The storage media 240 can include a hard disk, a floppy disk, a compact disc read-only memory ("CD-ROM"), a digital versatile disc ("DVD"), a Blu-ray disc, a magnetic tape, a flash memory, other non-volatile memory device, a solid state drive ("SSD"), any magnetic storage device, any optical storage device, any electrical storage device, any semiconductor storage device, any physical-based storage device, any other data storage device, or any combination or multiplicity thereof. The storage media 240 can store one or more operating systems, application programs and program modules, data, or any other information. The storage media 240 can be part of, or connected to, the computing machine. The storage media 240 can also be part of one or more other computing machines that are in communication with the computing machine such as servers, database servers, cloud storage, network attached storage, and so forth.

The applications module 300 can comprise one or more hardware or software elements configured to facilitate the computing machine with performing the various methods and processing functions presented herein. The applications module 300 can include one or more algorithms or sequences of instructions stored as software or firmware in association with the system memory 230, the storage media 240 or both. The storage media 240 can therefore represent examples of machine or computer readable media on which instructions or code can be stored for execution by the processor 210. Machine or computer readable media can generally refer to any medium or media used to provide instructions to the processor 210. Such machine or computer readable media associated with the applications module 300 can comprise a computer software product. It should be appreciated that a computer software product comprising the applications module 300 can also be associated with one or more processes or methods for delivering the applications module 300 to the computing machine 200 via a network, any signal-bearing medium, or any other communication or delivery technology. The applications module 300 can also comprise hardware circuits or information for configuring hardware circuits such as microcode or configuration information for an FPGA or other PLD. In one exemplary embodiment, applications module 200 can include algorithms capable of performing the functional operations described by the flow charts and computer systems presented herein.

The input/output ("I/O") interface 250 can be configured to couple to one or more external devices, to receive data from the one or more external devices, and to send data to the one or more external devices. Such external devices along with the various internal devices can also be known as peripheral devices. The I/O interface 250 can include both electrical and physical connections for coupling the various peripheral devices to the computing machine or the processor 210. The I/O interface 250 can be configured to communicate data, addresses, and control signals between the peripheral devices, the computing machine, or the processor 210. The I/O interface 250 can be configured to implement any standard interface, such as small computer system interface ("SCSI"), serial-attached SCSI ("SAS"), fiber channel, peripheral component interconnect ("PCI"), PCI express (PCIe), serial bus, parallel bus, advanced technology attached ("ATA"), serial ATA ("SATA"), universal serial bus ("USB"), Thunderbolt, FireWire, various video buses, and

the like. The I/O interface **250** can be configured to implement only one interface or bus technology. Alternatively, the I/O interface **250** can be configured to implement multiple interfaces or bus technologies. The I/O interface **250** can be configured as part of, all of, or to operate in conjunction with, the system bus **220**. The I/O interface **250** can include one or more buffers for buffering transmissions between one or more external devices, internal devices, the computing machine, or the processor **220**.

The I/O interface **220** can couple the computing machine to various input devices including mice, touch-screens, scanners, electronic digitizers, sensors, receivers, touchpads, trackballs, cameras, microphones, keyboards, any other pointing devices, or any combinations thereof. The I/O interface **220** can couple the computing machine to various output devices including video displays, speakers, printers, projectors, tactile feedback devices, automation control, robotic components, actuators, motors, fans, solenoids, valves, pumps, transmitters, signal emitters, lights, and so forth.

The computing machine **200** can operate in a networked environment using logical connections through the network interface **260** to one or more other systems or computing machines across a network. The network can include wide area networks (WAN), local area networks (LAN), intranets, the Internet, wireless access networks, wired networks, mobile networks, telephone networks, optical networks, or combinations thereof. The network can be packet switched, circuit switched, of any topology, and can use any communication protocol. Communication links within the network can involve various digital or an analog communication media such as fiber optic cables, free-space optics, waveguides, electrical conductors, wireless links, antennas, radio-frequency communications, and so forth.

The processor **210** can be connected to the other elements of the computing machine or the various peripherals discussed herein through the system bus **220**. It should be appreciated that the system bus **220** can be within the processor **210**, outside the processor **210**, or both. According to some embodiments, any of the processors **210**, the other elements of the computing machine, or the various peripherals discussed herein can be integrated into a single device such as a system on chip ("SOC"), system on package ("SOP"), or ASIC device.

Embodiments may comprise a computer program that embodies the functions described and illustrated herein, wherein the computer program is implemented in a computer system that comprises instructions stored in a machine-readable medium and a processor that executes the instructions. However, it should be apparent that there could be many different ways of implementing embodiments in computer programming, and the embodiments should not be construed as limited to any one set of computer program instructions unless otherwise disclosed for an exemplary embodiment. Further, a skilled programmer would be able to write such a computer program to implement an embodiment of the disclosed embodiments based on the appended flow charts, algorithms and associated description in the application text. Therefore, disclosure of a particular set of program code instructions is not considered necessary for an adequate understanding of how to make and use embodiments. Further, those skilled in the art will appreciate that one or more aspects of embodiments described herein may be performed by hardware, software, or a combination thereof, as may be embodied in one or more computing systems. Moreover, any reference to an act being performed

by a computer should not be construed as being performed by a single computer as more than one computer may perform the act.

The example embodiments described herein can be used with computer hardware and software that perform the methods and processing functions described previously. The systems, methods, and procedures described herein can be embodied in a programmable computer, computer-executable software, or digital circuitry. The software can be stored on computer-readable media. For example, computer-readable media can include a floppy disk, RAM, ROM, hard disk, removable media, flash memory, memory stick, optical media, magneto-optical media, CD-ROM, etc. Digital circuitry can include integrated circuits, gate arrays, building block logic, field programmable gate arrays (FPGA), etc.

The example systems, methods, and acts described in the embodiments presented previously are illustrative, and, in alternative embodiments, certain acts can be performed in a different order, in parallel with one another, omitted entirely, and/or combined between different example embodiments, and/or certain additional acts can be performed, without departing from the scope and spirit of various embodiments. Accordingly, such alternative embodiments are included in the description herein.

As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as "between X and Y" and "between about X and Y" should be interpreted to include X and Y. As used herein, phrases such as "between about X and Y" mean "between about X and about Y." As used herein, phrases such as "from about X to Y" mean "from about X to about Y."

As used herein, "hardware" can include a combination of discrete components, an integrated circuit, an application-specific integrated circuit, a field programmable gate array, or other suitable hardware. As used herein, "software" can include one or more objects, agents, threads, lines of code, subroutines, separate software applications, two or more lines of code or other suitable software structures operating in two or more software applications, on one or more processors (where a processor includes one or more microcomputers or other suitable data processing units, memory devices, input-output devices, displays, data input devices such as a keyboard or a mouse, peripherals such as printers and speakers, associated drivers, control cards, power sources, network devices, docking station devices, or other suitable devices operating under control of software systems in conjunction with the processor or other devices), or other suitable software structures. In one exemplary embodiment, software can include one or more lines of code or other suitable software structures operating in a general purpose software application, such as an operating system, and one or more lines of code or other suitable software structures operating in a specific purpose software application. As used herein, the term "couple" and its cognate terms, such as "couples" and "coupled," can include a physical connection (such as a copper conductor), a virtual connection (such as through randomly assigned memory locations of a data memory device), a logical connection (such as through

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logical gates of a semiconducting device), other suitable connections, or a suitable combination of such connections. The term "data" can refer to a suitable structure for using, conveying or storing data, such as a data field, a data buffer, a data message having the data value and sender/receiver address data, a control message having the data value and one or more operators that cause the receiving system or component to perform a function using the data, or other suitable hardware or software components for the electronic processing of data.

In general, a software system is a system that operates on a processor to perform predetermined functions in response to predetermined data fields. For example, a system can be defined by the function it performs and the data fields that it performs the function on. As used herein, a NAME system, where NAME is typically the name of the general function that is performed by the system, refers to a software system that is configured to operate on a processor and to perform the disclosed function on the disclosed data fields. Unless a specific algorithm is disclosed, then any suitable algorithm that would be known to one of skill in the art for performing the function using the associated data fields is contemplated as falling within the scope of the disclosure. For example, a message system that generates a message that includes a sender address field, a recipient address field and a message field would encompass software operating on a processor that can obtain the sender address field, recipient address field and message field from a suitable system or device of the processor, such as a buffer device or buffer system, can assemble the sender address field, recipient address field and message field into a suitable electronic message format (such as an electronic mail message, a TCP/IP message or any other suitable message format that has a sender address field, a recipient address field and message field), and can transmit the electronic message using electronic messaging systems and devices of the processor over a communications medium, such as a network. One of ordinary skill in the art would be able to provide the specific coding for a specific application based on the foregoing disclosure, which is intended to set forth exemplary embodiments of the present disclosure, and not to provide a tutorial for someone having less than ordinary skill in the art, such as someone who is unfamiliar with programming or processors in a suitable programming language. A specific algorithm for performing a function can be provided in a flow chart form or in other suitable formats, where the data fields and associated functions can be set forth in an exemplary order of operations, where the order can be rearranged as suitable and is not intended to be limiting unless explicitly stated to be limiting.

The above-disclosed embodiments have been presented for purposes of illustration and to enable one of ordinary skill in the art to practice the disclosure, but the disclosure is not intended to be exhaustive or limited to the forms disclosed. Many insubstantial modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The scope of the claims is intended to broadly cover the disclosed embodiments and any such modification. Further, the following clauses represent additional embodiments of the disclosure and should be considered within the scope of the disclosure:

Clause 1, an apparatus for generating forecasts from a high-dimensional parameter data space, the apparatus comprising: a reservoir model comprising input variables, output variables, and at least one algorithmic model, wherein the input variables, the output variables, and the at least one algorithmic model are

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generated by a flow simulator module and variables from at least one selected from a group comprising a formation and reservoir properties database and a field database; at least one selected from a group comprising a model order reduction module and an assisted history matching module; wherein the model order reduction module generates a subset of the input variables, the subset having a reduced parameter space, by identifying input variables that can be used to approximate at least one output variable; and wherein the assisted history matching module adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data;

Clause 2, the apparatus of clause 1, wherein the flow simulator module generates the reservoir model using at least one selected from a group comprising a full-physics flow simulator and a proxy flow simulator;

Clause 3, the apparatus of clause 1, wherein the model order reduction module uses a function decomposition algorithm or a design of experiments algorithm to identify input variables that can be used to approximate at least one output variable;

Clause 4, the apparatus of clause 1, wherein the input variables include reservoir parameters represented by a coefficient and the subset is a subset of the reservoir parameters represented by another coefficient having a value less than that of the coefficient;

Clause 5, the apparatus of clause 4, wherein the other coefficient is derived by applying a series expansion algorithm to the input variables and the output variables;

Clause 6, the apparatus of clause 5, further comprises a sensitivity analysis module, the sensitivity analysis module measures sensitivity of at least one output variable based on change in values of the subset of the reservoir parameters;

Clause 7, the apparatus of clause 1, wherein the assisted history matching module using Bayesian optimization to adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data;

Clause 8, a system for generating forecasts from a high-dimensional parameter data space, the apparatus comprising: at least one selected from a group comprising a full-physics flow simulator module and a proxy flow simulator module; at least one selected from a group comprising a model order reduction module and an assisted history matching module; wherein the full-physics flow simulator generates a reservoir model comprising input variables, output variables, and at least one algorithmic model, wherein the input variables, output variables, and the at least one algorithmic model are generated using variables from at least one selected from a group comprising a formation and reservoir properties database and a field database; wherein the proxy flow simulator generates a reservoir model comprising input variables, output variables, and at least one algorithmic model, wherein the input variables, output variables, and the at least one algorithmic model are generated using a machine learning algorithm, at least one selected from a group comprising a proxy flow simulator for assisted history matching and proxy flow simulator for field development optimization, and variables from at least one selected from a group comprising a formation and reservoir properties database and a field database; wherein the model

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order reduction module generates a subset of the input variables, the subset having a reduced parameter space, by identifying input variables that can be used to approximate at least one output variable; and wherein the assisted history matching module adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data;

Clause 9, the system of clause 8, wherein the flow simulator module generates the reservoir model using at least one selected from a group comprising a full-physics flow simulator and a proxy flow simulator;

Clause 10, the system of clause 8, wherein the model order reduction module uses a function decomposition algorithm or a design of experiments algorithm to identify input variables that can be used to approximate at least one output variable;

Clause 11, the system of clause 8, wherein the input variables include reservoir parameters represented by a coefficient and the subset is a subset of the reservoir parameters represented by another coefficient having a value less than that of the coefficient;

Clause 12, the system of clause 11, wherein the other coefficient is derived by applying a series expansion algorithm to the input variables and the output variables;

Clause 13, the system of clause 12, further comprises a sensitivity analysis module, the sensitivity analysis module measures sensitivity of at least one output variable based on change in values of the subset of the reservoir parameters;

Clause 14, the system of clause 8, wherein the assisted history matching module using Bayesian optimization to adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data;

Clause 15, a method for generating forecasts from a high-dimensional parameter data space, the apparatus comprising: generating a reservoir model, using a flow simulator, comprising input variables, output variables, and at least one algorithmic model using a flow simulator and variables from at least one selected from a group comprising a formation and reservoir properties database and a field database; generating, using model order reduction, a subset of the input variables, the subset having a reduced parameter space, by identifying input variables that can be used to approximate at least one output variable; and adjusting, using assisted history matching, values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data;

Clause 16, the method of clause 15, further comprising generating the reservoir model using at least one selected from a group comprising a full-physics flow simulator and a proxy flow simulator;

Clause 17, further comprising generating the subset of input variables by identifying, using either function decomposition or design of experiments, input variables that can be used to approximate at least one output variable;

Clause 18, the method of clause 15, further comprising: the input variables include reservoir parameters represented by a coefficient; and generating another coefficient having a coefficient less than that of the coefficient, wherein the other coefficient has a value less than that of the coefficient;

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Clause 19, the method of clause 18, further comprises: deriving the other coefficient by applying a series expansion algorithm to the input variables and the output variables; and measuring sensitivity of at least one output variable based on change in values of the subset of the reservoir parameters; and

Clause 20, the method of clause 15, further comprising applying, using assisted history matching, Bayesian optimization to adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data.

What is claimed is:

1. An apparatus for generating forecasts from an input high-dimensional parameter data space, the apparatus comprising:
 - a reservoir model that includes a plurality of input variables, a plurality of output variables, and at least one algorithmic model, wherein the plurality of input variables, the plurality of output variables, and the at least one algorithmic model are generated by a flow simulator module based on variables from a formation and reservoir properties database and a field database;
 - a model order reduction module configured to generate a subset of the plurality of input variables, the subset having a reduced parameter space relative to the input high-dimensional parameter data space by identifying input variables within the plurality of input variables useful to approximate at least one output variable of the plurality of output variables, wherein the model order reduction module uses a function decomposition algorithm to identify the input variables within the plurality of input variables useful to approximate the at least one output variable to generate the subset of the plurality of input variables with a reduced a number of input variables such that the subset of input variables of the plurality of input variables most effectively approximates the at least one output variable; and
 - an assisted history matching module configured to adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data.
2. The apparatus of claim 1, wherein the flow simulator module generates the reservoir model using at least one of a full-physics flow simulator and a proxy flow simulator.
3. The apparatus of claim 1, wherein the model order reduction module additionally uses a design of experiments algorithm to identify input variables that are useful to approximate at least one output variable.
4. The apparatus of claim 1, wherein the plurality of input variables include reservoir parameters represented by a first coefficient and the subset is a subset of the reservoir parameters represented by a second coefficient having a value less than that of the first coefficient.
5. The apparatus of claim 4, wherein the second coefficient is derived by applying a series expansion algorithm to the plurality of input variables and the plurality of output variables.
6. The apparatus of claim 5, further comprising a sensitivity analysis module configured to measure sensitivity of at least one output variable of the plurality of output variables based on change in values of the subset of the reservoir parameters.
7. The apparatus of claim 1, wherein the assisted history matching module uses Bayesian optimization to adjust values of at least one of the plurality of output variables based

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on a difference between the at least one of the output variables and dynamic field production data.

8. A system for generating forecasts from an input high-dimensional parameter data space, the system comprising:

at least one flow simulator configured to generate a reservoir model including a plurality of input variables, a plurality of output variables, and at least one algorithmic model, wherein the at least one flow simulator includes at least one of a full-physics flow simulator module and a proxy flow simulator module,

wherein the full-physics flow simulator module generates the reservoir model using variables from at least one of a formation and reservoir properties database and a field database,

wherein the proxy flow simulator module generates the reservoir model using a machine learning algorithm, an assisted history matching module and/or field development optimization module, and variables from at least one of the formation and reservoir properties database and the field database;

a model order reduction module configured to generate a subset of the plurality of input variables, the subset having a reduced parameter space relative to the input high-dimensional parameter data space by identifying input variables within the plurality of input variables useful to approximate at least one output variable of the plurality of output variables, wherein the model order reduction module uses a function decomposition algorithm to identify the input variables within the plurality of input variables useful that can be used to approximate the at least one output variable to generate the subset of the plurality of input variables with a reduced number of input variables such that the subset of input variables of the plurality of input variables most effectively approximates the at least one output variable; and an assisted history matching module configured to adjust values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data.

9. The system of claim 8, wherein the model order reduction module additionally uses a design of experiments algorithm to identify input variables that are useful to approximate at least one output variable.

10. The system of claim 8, wherein the plurality of input variables include reservoir parameters represented by a first coefficient and the subset is a subset of the reservoir parameters represented by a second coefficient having a value less than that of the first coefficient.

11. The system of claim 10, wherein the second coefficient is derived by applying a series expansion algorithm to the plurality of input variables and the plurality of output variables.

12. The system of claim 11, further comprising a sensitivity analysis module configured to measure sensitivity of at least one output variable of the plurality of output variables based on change in values of the subset of the reservoir parameters.

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13. The system of claim 8, wherein the assisted history matching module uses Bayesian optimization to adjust values of at least one of the plurality of output variables based on a difference between the at least one of the output variables and dynamic field production data.

14. A method for generating forecasts from an input high-dimensional parameter data space, the method comprising:

generating a reservoir model that includes a plurality of input variables, a plurality of output variables, and at least one algorithmic model, wherein the generation of the reservoir model is performed using a flow simulator and variables from at least one of a formation and reservoir properties database and a field database;

generating, using a model order reduction module, a subset of the plurality of input variables, the subset having a reduced parameter space relative to the input high-dimensional parameter data space by identifying input variables within the plurality of input variables useful to approximate at least one output variable of the plurality of output variables, wherein the model order reduction module uses a function decomposition algorithm to identify input variables within the plurality of input variables useful to approximate the at least one output variable to generate the subset of the plurality of input variables with a reduced number of input variables such that the subset of input variables of the plurality of input variables most effectively approximates the at least one output variable; and

adjusting, using an assisted history matching module, values of at least one of the output variables based on a difference between the at least one of the output variables and dynamic field production data.

15. The method of claim 14, wherein the flow simulator used for the generation of the reservoir model is at least one of a full-physics flow simulator and a proxy flow simulator.

16. The method of claim 14, wherein the generation of the subset of input variables is also performed using a design of experiments algorithm, input variables that are useful to approximate at least one output variable.

17. The method of claim 14, wherein the plurality of input variables include reservoir parameters represented by a first coefficient and subset is a subset of the reservoir parameters represented by a second coefficient having a value less than that of the first coefficient.

18. The method of claim 17, further comprising deriving the second coefficient by applying a series expansion algorithm to the input variables and the output variables, and measuring sensitivity of at least one output variable based on change in values of the subset of the reservoir parameters.

19. The method of claim 14, further comprising applying Bayesian optimization, using assisted history matching, to adjust values of at least one of the output variables based on a difference between the at least one of the plurality of output variables and dynamic field production data.

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